The Skyhook Tether

A Rotating Orbital Sling That Catches Cargo Low and Flings It High

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Abstract

Reaching orbit is only half the problem; the other half is climbing from a low orbit to a higher one, or onward to another world, and that too normally costs propellant. This study describes a way to make that climb without a rocket, by borrowing momentum from something already in orbit. The Skyhook Tether is a long, spinning cable in orbit: one end sweeps down low and fast toward a payload rising from below, catches it, carries it around, and releases it at the top of the swing at a much higher speed and altitude — a sling that lifts cargo by trading its own orbital momentum. Because the tether gives up momentum each time it throws, it slows and drops, but it can be restored electrically or by catching returning traffic, so that over many exchanges no propellant need be spent at all. The physics is elementary orbital mechanics, long studied under the name of momentum-exchange tethers, and it pairs naturally with the mass driver of the companion concept, which throws cargo up to where the skyhook can catch it. We describe the mechanism, the momentum bookkeeping, the honest problem of catching a payload precisely, and its place in a propellant-free transport chain.

1. Introduction: The Second Half of the Climb

Getting off a world's surface is the hardest and most celebrated part of spaceflight, but it is not the whole journey. A payload that reaches a low orbit is still deep in a gravity well relative to where it often needs to be: a higher orbit, an escape trajectory, a transfer to another world. Each of those steps requires a further increase in speed, and each is normally paid for with propellant carried up at great cost from below. The mass driver of the companion concept, and rockets generally, can put cargo into a low orbit; the question this concept answers is how to lift it from there without burning yet more fuel.

The answer is to take the needed momentum from something that already has it in abundance: a large object in orbit. Orbital motion is a vast reservoir of momentum, and if a payload can be coupled briefly to a massive orbiting structure and then released, momentum can be transferred from the structure to the payload — flinging the payload higher and faster while the structure gives up a little of its own motion. A rotating tether does exactly this, and does it as a graceful mechanical throw. The skyhook is the sling that catches cargo in a low orbit and hurls it toward a high one, powered not by fuel but by the momentum of orbit itself.

2. Concept Description

The Skyhook Tether is a long cable in orbit, with a massive station or counterweight at its centre, set spinning end over end as it orbits. The rotation is arranged so that at the lowest point of its swing, one end of the tether dips down and momentarily moves backward relative to its orbital motion — that is, more slowly over the ground — reaching low and slow enough to meet a payload climbing up from below. At that instant the payload is caught and attached to the tether tip. The tether carries it around and up, and at the top of the swing, half a rotation later, the tip is moving forward and fast; there the payload is released, flung away at a much greater speed and altitude than it arrived with. In effect the tether reaches down, grabs the cargo at a convenient low speed, and slings it upward and onward.

The elegance is that the catch happens where the tether tip is moving slowly relative to the incoming payload, so the two need not match the full orbital speed difference by rocket — the tether's rotation cancels much of it — and the release happens where the tip is moving fast, adding that speed to the payload for free. A single well-designed rotating tether can take a payload from a low orbit and throw it onto a transfer toward the Moon, another planet, or a higher orbit, entirely by mechanical exchange of momentum, with no propellant expended in the throw (Moravec, 1977, on the rotating skyhook concept).

3. The Physics: Trading Momentum, Not Burning Fuel

A rotating tether in orbit has two motions combined: it orbits the planet, and it spins about its own centre. At the bottom of each spin the tip's rotational motion points backward, subtracting from the orbital speed, so the tip briefly moves slowly over the ground; at the top the rotational motion points forward, adding to the orbital speed, so the tip moves fast. A payload caught at the slow bottom and released at the fast top gains the full swing in speed — twice the tip's rotational velocity — lifted from a low, slow state to a high, fast one without any engine firing on the payload. This is momentum exchange: the payload gains momentum, and by conservation the tether loses an equal amount (Moravec, 1977).

That loss is the crux of the bookkeeping. Each throw slows the tether and lowers its orbit slightly, and if nothing replaced that momentum the tether would eventually spiral down. But the momentum can be restored without propellant in two ways: electrically, using a conducting tether pushing against the planet's magnetic field to raise its orbit over time, or by symmetry, catching payloads coming the other way — descending traffic that gives momentum back to the tether as it is lowered gently toward the surface. A tether that balances upward and downward traffic, or that slowly recharges its orbit electrically, can operate indefinitely as a genuinely propellant-free lift, trading momentum in a closed economy rather than consuming fuel (Hoyt and Uphoff, 2000, on tether momentum management and cislunar transport).

4. The Machine and Its Partner

A skyhook comprises a long, strong tether, a central station or counterweight housing the systems, a capture mechanism at the tether tip able to grab an incoming payload cleanly and quickly, and a means of restoring orbital momentum — an electrodynamic system, a balance of two-way traffic, or both. Because the tether must survive the tension of spinning with a payload on its tip and the hazard of small impacts over a long life, it is best built not as a single strand but as a redundant, multi-strand structure that cannot be severed by a single cut (Hoyt and Uphoff, 2000). The whole assembly rotates in a carefully controlled way so that the tip arrives at the catch point on time, at the right place, and at the right slow speed.

The skyhook is the natural partner to the mass driver. A launcher on a moon or small world throws rugged cargo up to a low altitude but cannot easily give it the further boost to a transfer orbit; a skyhook waiting above can catch that cargo at the top of its arc and fling it onward. Together they form a propellant-free chain: electricity accelerates cargo off the surface, and orbital momentum lifts it the rest of the way, with fuel spent nowhere in the sequence. The skyhook can equally serve worlds with atmospheres, catching payloads brought up by a suborbital rocket or spaceplane to a modest height and speed, and completing their climb to orbit — halving the velocity a launch vehicle must supply, and so transforming its economics.

Step	What happens	Speed / energy
Catch (bottom)	tip dips low, moving slowly	low relative speed
Carry	payload swung around + up	no fuel used
Release (top)	tip fast, flings payload	high speed gained
Tether pays	loses momentum, drops a little	restored later
Restore	electrodynamic / return traffic	no propellant

Table 1. The skyhook's cycle — momentum in, momentum out, no fuel burned.

5. Operations Concept

Phase one is a demonstrator tether that catches and throws a small, instrumented payload, proving the capture, the momentum exchange, and the restoration of orbit at low stakes. Phase two is an operational skyhook paired with a surface launcher — the mass driver on a moon, or a suborbital vehicle on a world with air — catching cargo delivered to a low rendezvous and flinging it to a transfer orbit, so that the propellant cost of the second half of the climb is eliminated. Phase three is a network of tethers exchanging traffic across cislunar space and beyond, catching and releasing payloads between orbits and worlds, moving mass through the solar system on borrowed momentum rather than burned fuel. Throughout, the skyhook's value is greatest where it closes a chain: paired with a launcher below and a catcher above, it becomes a link in a transport system that spends no propellant at all.

6. Honest Difficulties

The defining difficulty is the catch. Capturing a payload at the tether's tip demands that the payload and the tip meet at exactly the right place, time, and relative speed, with the tether rotating and orbiting and the payload climbing to meet it — a rendezvous of unforgiving precision, and a failed catch is a serious event. The tether itself must be strong enough to swing a payload without breaking, which for large payloads demands very strong materials and clever multi-strand designs to resist both tension and impacts over a long life (Hoyt and Uphoff, 2000). Managing the tether's momentum so that it neither spirals down nor drifts up requires an effective restoration system and careful scheduling of traffic. The rotating dynamics are intricate, and control must be precise. None of this is a barrier of physics — momentum-exchange tethers are a thoroughly analysed idea (Moravec, 1977; Hoyt and Uphoff, 2000) — but the precision of capture and the strength of the tether are real engineering hurdles, and honesty places the catch first among them.

7. Pathway to Reality

The concept rests on well-understood physics and decades of analysis. Momentum-exchange and rotating tethers have been studied in detail since the 1970s (Moravec, 1977), tether survivability and momentum management have been worked out (Hoyt and Uphoff, 2000), and space tethers of various kinds have been flown in orbit, demonstrating deployment and electrodynamic effects. The credible near-term step is a demonstrator that performs a real catch-and-throw with a small payload and shows that the orbit can be restored, before scaling to operational mass. Because the skyhook multiplies the value of the mass driver and of any modest launcher, completing the climb that surface launch begins, it is a keystone of a mature, propellant-thrifty transport network — less a vehicle than a piece of orbital infrastructure that, once in place, lifts everything that follows.

8. Conclusion

A rocket climbs by throwing part of itself away; a skyhook climbs by borrowing speed and giving it back later, so that nothing is thrown away at all. It reaches down into a low orbit, catches a rising payload at a gentle relative speed, and slings it upward on the strength of its own orbital motion, paying the momentum forward and recovering it from returning traffic or from the planet's magnetic field. The catch is delicate and the tether must be strong, and it is worth building only as part of a chain — but in that chain, paired with a launcher below, it turns the punishing second half of the climb into a free ride on momentum that was already there. It is the orbital sling that makes the solar system smaller, one silent, fuel-less throw at a time.

References

Harvard referencing style (Cite Them Right).

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